

2 Complexity Analysis

The overall computational complexity of MDPRec is governed by the sequence length L and the embedding dimension D . The embedding layer, consisting of item/position lookups and the hierarchical timestamp encoding, costs $\mathcal{O}(LD)$. For each temporal scale branch, the Coarse-grained MoE and Fine-grained MoE compute gating scores via a linear layer and softmax, which requires $\mathcal{O}(LDK_c)$ and $\mathcal{O}(LDK_f)$ operations, respectively, with the expert aggregation adding a negligible $\mathcal{O}(LD)$ term. The each branch is the period-aware sequence encoder built upon NormAtt. In NormAtt, a learned weight matrix $\mathbf{W} \in \mathbb{R}^{L \times L}$ is multiplied by the input sequence, incurring $\mathcal{O}(L^2D)$ cost per layer; the differentiable periodic modulator $\mathcal{S}(\gamma; \alpha, \beta)$ constructs the mask $\mathbf{M}$ by evaluating pairwise distances between all time steps, which takes $\mathcal{O}(L^2)$ additional operations. The feed-forward network contributes $\mathcal{O}(LD^2)$. Stacking n such NormAtt-FFN layers in each branch therefore yields a complexity of $\mathcal{O}(n(L^2D + LD^2))$. The Dual Cross-Scale Dependency Interaction MoE concatenates the outputs of the two branches, computes gating weights, and applies K_d expert transformations, leading to $\mathcal{O}(K_dLD^2)$ cost. Finally, the gated fusion and dot-product prediction against the item embedding matrix introduce $\mathcal{O}(LD + |J|D)$ operations. In summary, the total computational complexity of MDPRec is $\mathcal{O}(nL^2D + (n + K_d)LD^2 + |J|D)$. Because the sequence length L is fixed to a moderate value (e.g., 50 in our experiments), the quadratic term remains manageable, making MDPRec both expressive and practically efficient.